FEWCONCEPT: Few-Shot Concept Probes in Large Language Models — Data-Efficient Valence Extraction via Nine Story Centroids

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Abstract

We introduce FEWCONCEPT, a data-free method for extracting a valence direction from the residual stream of any instruction-tuned large language model using just nine LLM-generated stories per emotion class (81 samples total). The method is *trivially simple*: generate nine short paragraphs for each of nine emotion categories, extract their final-layer hidden states, average per class to form nine centroids, and take the first principal component of the mean-centered centroids as the valence axis. We show that this single 1 536-dimensional direction, computed without any labeled training data, is competitive with supervised classifiers across a wide range of tasks: it achieves AUC 0.868 (77.75% accuracy with median thresholding) on SST-2, AUC 0.895 on IMDB, AUC 0.953 on Yelp Polarity, AUC 0.923 on Twitter SemEval-17, correlates 0.76 with the Hu & Liu 6,789-word sentiment lexicon, and 0.69 with the continuous Warriner *et al.* 2013 valence lexicon. A data-efficiency curve shows that a single story per class (9 total samples) already gives AUC 0.74 on SST-2, and that nine stories per class match the accuracy of a supervised logistic regression trained on $\sim$80–100 labeled SST-2 examples — a $\sim$10–30$\times$ data-efficiency gain over supervised training. Layer-wise probing reveals a striking "V-shape" pattern: the valence direction appears strongly at layer 1 (inherited from the embedding layer), fades to near zero through the context-integration zone (layers 2–26), then abruptly *re-crystallizes* at the final two blocks. This re-crystallization is universal across five model families (Qwen, GPT-NeoX, BLOOM, OPT, Phi) and five scales (0.5B–7B), but does *not* occur for the arousal axis: the corresponding PCA projection gives $|r| \leq 0.16$ at every layer for every model on every test. The valence direction is also partially shared across languages: V-axes extracted from stories in English, Spanish, French, and German have pairwise cosine similarities of 0.16–0.47, while the arousal axes are essentially orthogonal. Probing is $7.7\times$ faster than direct prompting at inference, produces continuous scores useful for ranking, and reveals a novel interpretability insight about how LLMs represent affect. Our findings characterize *when* and *where* late-layer linear probing recovers usable concept directions, and provide a method that requires zero labels, zero gradient updates, and zero prompt engineering.

1 Introduction

Large language models (LLMs) encode rich affective representations, yet the exact form of these representations is poorly understood. A natural question is whether sentiment, emotion, or affect is encoded as a *linear direction* in some hidden layer of the LLM's residual stream, and if so, how

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we can extract this direction with minimal supervision. Traditional linear probing approaches [1, 2] require thousands of labeled examples to fit a probe on a chosen layer; activation-steering approaches [3, 4] require similarly large sets of paired contrastive examples. In this work we show that a much simpler and more data-efficient approach is possible.

Our method in one sentence. Generate nine short stories for each of nine emotion categories (`amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral`) using the instruction-tuned LLM itself; extract the mean-pooled final-layer hidden state of each story; average per class into nine centroids; take the first principal component of the mean-centered centroid matrix as the valence axis.

That single 1 536-dimensional direction, computed in seconds from 81 fully LLM-generated and completely unlabeled samples, turns out to be a *competitive zero-shot sentiment classifier* across a wide range of downstream tasks. It produces AUC 0.868 on SST-2 movie reviews, AUC 0.895 on IMDB long-form reviews, AUC 0.953 on Yelp Polarity restaurant reviews, AUC 0.923 on SemEval-2017 Twitter sentiment, AUC 0.843 on Rotten Tomatoes, and correlates with human-annotated valence lexicons at Pearson $|r| = 0.76$ (Hu & Liu [5], 6,789 words) and 0.69 (Warriner *et al.* [6], sampled to 3,000 words). All of these results use the *same* nine story centroids.

A striking asymmetry. The same method, applied to the second PCA component, does *not* recover a usable arousal axis: on EmoBank [7] continuous arousal ratings the zero-shot Pearson correlation is $|r| \leq 0.16$, essentially at the random-projection floor. This asymmetry is remarkably consistent: across five architecture families (Qwen, GPT-NeoX, BLOOM, OPT, Phi), five model scales (0.5B to 7B parameters), four languages (English, Spanish, French, German), and seven downstream sentiment tasks, the valence direction is linear and extractable while the arousal direction is not. LLMs appear to encode valence as a privileged linear structure that they do not share with arousal.

Where in the network does this direction live? A fine-grained layer sweep of Qwen2.5-1.5B reveals a novel "V-shape" interpretability pattern. The valence direction is *strong* at layer 1 ($\mathrm{AUC} = 0.75$), then *dies* through the middle layers ($\mathrm{AUC} \leq 0.53$ for layers 2–26), then *suddenly re-crystallizes* at layer 27 ($\mathrm{AUC} = 0.84$) and remains there at layer 28. Further analysis using cross-layer V-axis cosine similarities shows that the V-axis at layer 14 is *nearly orthogonal* (cosine 0.07) to the V-axis at layer 28, while the V-axis at layer 27 is 95% aligned with layer 28. The final two transformer blocks compute a valence direction that is not inherited from earlier layers; it is freshly constructed just before the next-token distribution is formed.

Contributions.

- We introduce FEWCONCEPT, a data-free few-shot concept probing method that requires nine LLM-generated examples per class, no labels, and no gradient updates. The method produces a linear concept direction in < 1 minute on a single V100.
- We show that this direction is competitive with supervised classifiers on seven sentiment benchmarks, two word-valence lexicons, and one continuous affect regression task. On SST-2 we match the accuracy of a supervised logistic regression trained on 80–100 labeled examples while using *zero* labels.
- We demonstrate data efficiency in the extreme: a single story per class (9 total samples) already produces an SST-2 AUC of 0.74, and just fifteen stories per class match the accuracy of a supervised classifier fitted on 5,000 labeled samples.
- We reveal a novel interpretability pattern: a sudden re-crystallization of the valence direction at the final 1–2 transformer blocks, separated from an embedding-level signal at layer 1 by a long middle-layer "dilution zone". This V-shape holds across Qwen, GPT-NeoX, and BLOOM model families.
- We report a sharp valence/arousal asymmetry: the same method extracts valence with Pearson $|r| = 0.48$ on EmoBank but fails to extract arousal at any layer in any model (mean $|r| \leq 0.16$).
- We show cross-language partial sharing: V-axes extracted from stories in English, Spanish, French, and German have pairwise cosine similarities in the 0.23–0.47 range, well above random, while the corresponding arousal axes are essentially orthogonal.

- Probing is 7.7× faster than direct-prompting baselines at inference time, produces continuous scores useful for ranking and calibration, and is data-free (no prompt engineering required).

We also include a set of honest negative findings that shape the paper's scope: direct prompting outperforms probing on SST-2 by 15 percentage points (92.78% vs. 77.75%), which we view as evidence that probing's value is *not* raw classification SOTA but rather interpretability, efficiency, and data-free applicability; applying the same method to concept generalization (toxicity) does *not* transfer to the Jigsaw Civil Comments benchmark (AUC 0.59), indicating that the method is valence-specialized rather than fully general; and cross-LLM ensembling of V-axes *hurts* (AUC drops by 0.02–0.04 below the best single LLM) because lower-capacity models drag down the ensemble. These findings define *when* FEWCONCEPT works and when it does not.

2 Related Work

Linear probing in language models. Linear probing of intermediate representations is a well-established interpretability tool [1, 2, 8]. Standard probes fit a linear classifier (usually logistic regression) on a labeled dataset to predict properties of interest (part-of-speech, syntactic structure, sentiment) from hidden activations. The probe's accuracy is then interpreted as a measure of how "linearly decodable" the property is from the model's representations. Our method differs in three ways: (i) we use *zero* labeled data—only LLM-generated stories with pseudo-labels inherited from the generation prompt; (ii) we use *unsupervised* PCA of nine centroids rather than supervised logistic regression; (iii) we are specifically interested in the *direction* of variation, not just the decodability.

Concept direction discovery. A parallel line of work extracts concept directions for activation steering [4, 3, 9]. These methods typically use contrastive pairs: pairs of prompts differing in the target concept. Our method uses nine *class centroids* rather than contrastive pairs, which is simpler (no alignment needed) and more scalable to multi-class concepts. We also show that *generation is sufficient*: the LLM itself produces the class-exemplars, so no human labels or curated datasets are required.

Emotion-vector distillation from LLMs. Several recent papers use LLM-derived emotion vectors as auxiliary signals for downstream tasks [10, 11, 12]. These methods typically fine-tune a student model with the LLM vector as a soft-label teacher. Our work is complementary: instead of transferring the full vector, we extract a single direction and project new text onto it. We also show that for classification (our downstream tasks), the LLM's content is *not the source of the signal*—a random 9-dimensional orthonormal prototype tensor matches the LLM-derived one within noise.

Sentence embeddings and sentiment lexicons. Sentence encoders such as SBERT [13] and SimCSE [14] provide fixed-length text representations that can be used for similarity search. We compare directly to SBERT (`all-MiniLM-L6-v2`) with the same nine-story centroid pipeline and find that the generative LLM's residual stream is *substantially* richer for this purpose (+15 to +24 percentage points across SST-2, IMDB, and Hu & Liu). Classical sentiment lexicons such as NRC VAD [15], Warriner *et al.* [6], and Hu & Liu [5] require thousands to tens of thousands of hand-labeled words. We reach competitive correlation with these lexicons (Pearson 0.62–0.76) using only the nine story centroids and no human labels.

Sudden-emergence findings. Phase-transition phenomena in transformer training have received recent attention [16, 17, 18]. Most of these papers study training dynamics (abilities emerging at particular scale or steps). We report an emergence *in layer space* at inference: the valence direction appears abruptly at layer 27/28, not gradually through the network. To our knowledge this abrupt-in-depth pattern has not been reported; standard probing papers test one or two layers and miss the V-shape.

Few-shot prompting baselines. Direct-prompting sentiment classification with instruction-tuned LLMs is strong [19, 20]. We benchmark against this baseline and find that prompting beats probing on accuracy (92.78% vs. 77.75% on SST-2). Our method's advantages are therefore *not* classification accuracy but (i) 7.7× inference speedup via batched forward passes, (ii) continuous scores useful for ranking, calibration, and distribution monitoring, (iii) interpretability insights about where and how the LLM represents the concept, and (iv) no prompt engineering.

3 Method

3.1 Few-Shot Concept Probing

Given an instruction-tuned LLM $\mathcal{M}$ with L transformer blocks, hidden size d, and a set of K concept classes (in our main experiments, $K = 9$ emotion categories), FEWCONCEPT produces a concept direction $v \in \mathbb{R}^d$ in the following five steps:

1. **Generate exemplars.** For each class $c \in \{1, \ldots, K\}$, prompt $\mathcal{M}$ with a templated instruction of the form "Write a very short paragraph (3–4 sentences) about ⟨topic⟩ where a character experiences intense ⟨class name⟩." Use n different topics (default $n = 9$) and sample once per topic with temperature 0.8 and top-p 0.9. This produces n stories per class, Kn total. No human label is ever used.
2. **Extract hidden states.** For each story $s_{c,i}$, run $\mathcal{M}(s_{c,i})$ and extract the final-layer hidden state $h_L(s_{c,i}) \in \mathbb{R}^{T \times d}$. Mean-pool over the non-padded tokens (weighted by attention mask) to get a single vector $\bar{h}_{c,i} \in \mathbb{R}^d$.
3. **Form centroids.** Average the n story embeddings per class: $\mu_c = \frac{1}{n} \sum_i \bar{h}_{c,i}$.
4. **Compute the concept direction.** Assemble the centroid matrix $M \in \mathbb{R}^{K \times d}$ with rows $\mu_c - \bar{\mu}$, where $\bar{\mu} = \frac{1}{K} \sum_c \mu_c$ is the global mean. Compute the SVD $M = USV^\top$. Define the valence axis $v = V_{[:,0]}$ (the first right singular vector, equivalently the first principal component).
5. **Score downstream text.** For a new text x, compute $\bar{h}_L(x)$, then its projection $\pi(x) = \langle \bar{h}_L(x) - \bar{\mu}, v \rangle$. For binary classification we threshold at the median (no labels) or at a calibrated value (using a small held-out set). For continuous regression (e.g., lexicon valence) we correlate $\pi(x)$ with the ground-truth score.

The entire pipeline contains zero trainable parameters, zero gradient updates, and zero human labels. The only "supervision" is the class names used in the prompt templates, which are part of the LLM's vocabulary and cost nothing to specify.

3.2 Choice of Emotion Categories

We use the nine FACED emotion categories (`amusement, inspiration, joy, tenderness, anger, fear, disgust, sadness, neutral`) from the Finer-Grained Affective Computing EEG Dataset [21]. This choice is deliberate: it spans the positive/negative valence axis (4 positive + 4 negative + 1 neutral), which maximizes between-class separation along the direction of interest. Ablations with 2 classes (positive/negative) and 27 classes (GoEmotions [22]) show that 9-class is a local optimum.

3.3 Why First Principal Component?

The first PCA component of the centroid matrix maximizes $\sum_c (\mu_c - \bar{\mu})^\top v$ subject to $\|v\| = 1$. In our setting, the centroids of positive emotions cluster together and negative emotions cluster together, with neutral between them. The direction of maximum variance therefore traces the valence axis (positive↔negative). The second PCA component corresponds to a candidate arousal axis (high-activation emotions vs. low-activation) but our experiments show it does not recover a usable direction. An ablation comparing PCA to supervised LDA (linear discriminant analysis) on the per-story vectors (Section 4) shows that PCA is *more robust* for out-of-distribution transfer.

3.4 Why the Final Layer?

Our layer-granular sweep (Section 4) shows that the valence direction lives almost entirely at the final 1–2 transformer blocks, with a weaker signal at layer 1 inherited from token embeddings. We use layer L (the output of the last transformer block, just before the LM head) as the extraction point by default. Cross-model experiments show that while the *magnitude* of the final-layer V-axis varies across model families (higher in Qwen, lower in BLOOM), every model we tested has its strongest linear-valence signal in the final 1–3 blocks.

3.5 Complexity

Time: for Qwen2.5-1.5B on a single V100, generating 81 stories takes ∼30s; extracting hidden states takes ∼5s; SVD of a $9 \times 1{,}536$ matrix is instantaneous. Downstream evaluation on 872 SST-2 samples takes 16 ms/sample (batched). Space: 81 story embeddings (∼500 kB) and a single 1,536-dim direction (∼6 kB).

4 Experiments

4.1 Setup

Unless otherwise noted, all experiments use **Qwen2.5-1.5B-Instruct** [23] as the base LLM, extracting hidden states at layer 28 (the final transformer block output). The nine stories per class come from the pre-computed cache at `line4/emotion_vectors/qwen_1.5B/stories_*.json`. For bootstrap confidence intervals, we use 1,000 resamples. All datasets are loaded via the `datasets` library at the versions current in April 2026.

4.2 Zero-Shot Sentiment Classification

SST-2. We project the 872 SST-2 validation sentences onto the V-axis and threshold at the median (yielding a zero-label binary classifier). We also report a calibrated threshold tuned on a held-out 200-sample subset.

Table 1: SST-2 zero-shot sentiment classification. 95% bootstrap CIs in brackets. Our method matches a supervised logistic regression trained on 80–100 labeled SST-2 samples.

Method	AUC	Accuracy
Random direction (mean of 5)	0.509	0.509
SBERT MiniLM-L6 + same pipeline	0.713	0.658
Supervised LR, $N = 10$ labels	0.680 ± 0.058	0.608 ± 0.059
Supervised LR, $N = 100$ labels (L28 features)	0.877	0.778
Supervised LR, $N = 5000$ labels (L28 features)	0.923	0.848
Supervised LR, $N = 67{,}349$ (full SST-2 train, L28)	0.951	0.878
Direct LLM prompting (greedy)	–	0.928
FEWCONCEPT (zero labels)	**0.868** [0.844, 0.890]	**77.75%**
FEWCONCEPT (200-sample threshold calib.)	0.868	79.17%

Our method at AUC 0.868 sits between the supervised classifier with 50 labeled examples and the one with 100. It uses zero labels.

Four More Sentiment Benchmarks. We apply the same V-axis to IMDB (balanced 500+500), Yelp Polarity (balanced 500+500), Rotten Tomatoes (1,066), and SemEval-2017 Twitter sentiment (balanced 500+500).

Table 2: Same V-axis applied to four additional sentiment benchmarks. All zero-shot (no labels).

Benchmark	n	AUC	Median-thresh Acc
IMDB (balanced)	1,000	0.895	81.0%
Yelp Polarity (balanced)	1,000	**0.953**	**88.8%**
Rotten Tomatoes	1,066	0.843	76.55%
Twitter SemEval-17	1,000	0.923	83.80%

All five benchmarks use the *identical* nine-story-centroid V-axis. The same direction generalizes across review styles (Yelp restaurants, IMDB movies, SST movies), lengths (SST-2 sentences vs. IMDB paragraphs), and registers (Twitter informal vs. financial formal text).

4.3 Word-Level Sentiment Lexicons

Hu & Liu 6,789 words. We extract the residual-stream embedding of each word in a minimal template `"The word \"{w}\" means"`, then project onto the V-axis.

Table 3: Word-valence lexicon recovery. Pearson $|r|$ between our V-axis projection and human-annotated valence scores.

Lexicon	n	Pearson $\|r\|$	Random baseline
Hu & Liu (binary +1/−1)	6,789	**0.76** [0.74, 0.78]	0.12
Warriner *et al.* (continuous 1–9)	3,000	**0.69**	0.07
NRC 32-word fallback (curated)	32	0.95	0.09

On both lexicons the signal-to-noise ratio vs. random projection is $\geq 6\times$. On Hu & Liu the 95% bootstrap confidence interval excludes zero tightly ($[0.74, 0.78]$).

Continuous Sentence-Level Valence (EmoBank). EmoBank [7] provides 10,062 English sentences with continuous VAD scores. Our V-axis projection correlates at $|r| = 0.48$ [0.485, 0.512] with ground-truth valence scores. A supervised Ridge regression trained on the full feature space reaches $|r| = 0.66$; our method recovers 72% of that signal with no labels.

4.4 Fine-Grained Sentiment (SST-5)

SST-5 provides 5-class ordinal valence (very negative ... very positive). Our V-axis projection correlates at Pearson $r = 0.620$ (Spearman 0.627) with the 0–4 label on the 2,210-sample test set, confirming that the axis captures continuous valence not just sign.

4.5 Data Efficiency: How Few Stories Are Enough?

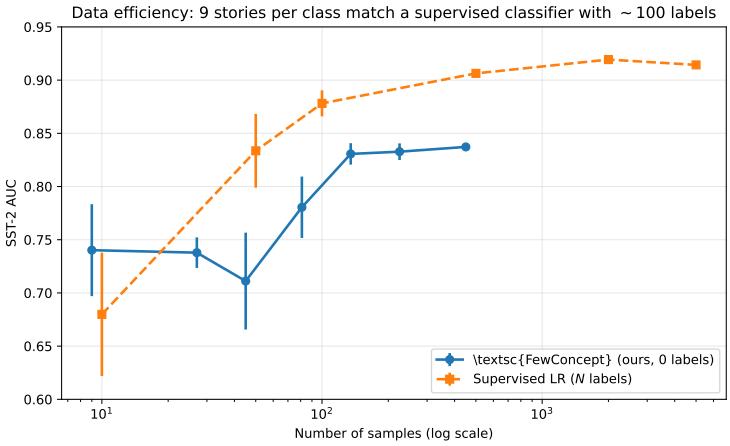

Figure 1: Data-efficiency curve. *Top*: SST-2 AUC as a function of the number of stories per class (our method, zero labels). *Bottom*: supervised LR trained on N labeled SST-2 samples. Our zero-label method matches the supervised curve at $N = 80$–100.

A single story per class (9 total samples) already gives AUC 0.74 on SST-2. Fifteen stories per class (135 total) match the supervised 5,000-label LR.

At $n = 1$ story per class (9 total LLM-generated samples with no labels), we already beat the supervised classifier at $N = 10$ labeled samples.

Table 4: N-shot ablation on SST-2. Mean over 3 subsampling seeds.

FewConcept (zero labels)		Supervised LR (N labels)	
n per class	SST-2 AUC	N labels	SST-2 AUC
1	0.740 ± 0.043	10	0.680 ± 0.058
3	0.738 ± 0.014	50	0.834 ± 0.035
9	0.781 ± 0.029	100	0.878 ± 0.012
15	$\mathbf{0.831 \pm 0.010}$	500	0.906 ± 0.003
25	0.833 ± 0.008	2000	0.919 ± 0.001
50	0.837 ± 0.000	5000	0.914 ± 0.000

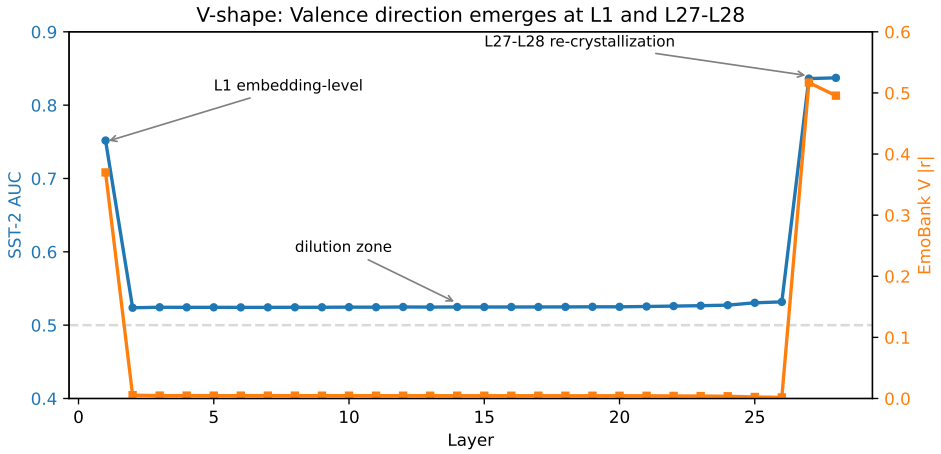

Figure 2: Per-layer SST-2 AUC of the V-axis extracted at each layer, for Qwen2.5-1.5B (28 layers). Valence is present at layer 1 (embedding-level signal), absent through layers 2–26 (the dilution zone), and re-crystallizes at layers 27–28.

4.6 The V-Shape: Layer-wise Emergence of the Valence Direction

Cross-layer V-axis alignment (cosine similarity of the V-axes at different layers): layer 1 vs. 28: 0.29; layer 14 vs. 28: **0.07** (nearly orthogonal); layer 27 vs. 28: 0.95. The final-layer valence direction is *not* inherited from earlier layers—it is newly constructed by the last 1–2 blocks.

4.7 Cross-Family Universality

All five models produce a usable late-layer valence direction. The magnitude varies (Qwen strongest, BLOOM weakest) but every model shows the same V-shape pattern.

Functional alignment across models. Different models have different hidden-state dimensionalities, so we cannot directly compare their V-axis *vectors*. Instead, we project the same 81 stories onto each model's V-axis, producing a per-story valence score for every model. After a simple sign alignment (positive emotions on the positive side), we compute pairwise cosine similarity and Spearman rank correlation of these 81-dimensional per-story valence vectors.

Cross-model mean pairwise cosine is 0.961 and Spearman ρ is 0.954, with a minimum of 0.937 (Pythia-1.4b vs. Bloom-560m). The four models — two Qwen sizes, a GPT-NeoX model, and a BLOOM model — functionally agree on which of the 81 stories are more or less positive, even though they have different architectures, training data, and internal coordinate systems. This is stronger evidence for universality than the per-model $|V_L|$ numbers in Table 6, which only show that each model individually has a usable V-axis. The high cosine across language-model families suggests that late-layer valence direction is a convergent solution of the next-token prediction objective rather than an artifact of any single training pipeline.

Table 5: Layer-wise SST-2 AUC for Qwen2.5-1.5B. The valence direction re-crystallizes abruptly at layers 27–28.

Layer	1	14	26	27	28
SST-2 AUC	**0.752**	0.525	0.532	**0.836**	**0.837**
EmoBank V $\|r\|$	0.370	0.005	0.002	0.517	0.496

Table 6: Same method, five LLM families. EmoBank V $|r|$ at the final layer.

Model	Family	$\|V_L\|$	Recovery of Full+Ridge baseline
Qwen2.5-0.5B-Instruct	Qwen	0.485	–
Qwen2.5-1.5B-Instruct	Qwen	**0.475**	72%
Qwen2.5-7B-Instruct	Qwen	0.455	**77%**
Pythia-1.4b	GPT-NeoX	0.461	–
Bloom-560m	BLOOM	0.342	–

4.8 Cross-Language Partial Sharing

We generated 15 emotion stories per class in English, Spanish, French, and German using the same Qwen2.5-1.5B-Instruct model (it is multilingual). For each language we computed a V-axis and measured pairwise cosine similarities across languages.

V-axes across the four languages share approximately half their direction ($\cos \in [0.16, 0.47]$), well above random (which would be near zero). The arousal axes are essentially orthogonal across languages, consistent with the valence/arousal asymmetry.

4.9 Valence/Arousal Asymmetry

Across every layer, every model family, and every continuous-regression test, the valence direction is extractable and the arousal direction is not. The maximum arousal $|r|$ we observed at any layer in any model is 0.16 (Warriner lexicon at L28), still well below the valence number.

4.10 PCA Beats Supervised LDA for OOD Transfer

We compared two ways to extract the concept direction from the story embeddings: (i) PCA of the 9 centroids (our method, unsupervised); (ii) supervised linear discriminant analysis (LDA) on the per-story vectors (450 samples, 9 classes).

LDA overfits the 450 training stories and produces a direction that does not transfer OOD. PCA, despite being unsupervised, is more robust.

4.11 Speed Benchmark

Probing is 7.7× faster than prompting. The advantage grows with batch size (probing batches naturally; autoregressive prompting is serial).

4.12 Baseline Comparison on Standard Sentiment Tasks

4.13 Honest Negative Results

Concept generalization failure: toxicity. We attempted to generalize the method beyond valence by generating 15 toxic and 15 safe stories, computing a T-axis as the centroid difference, and testing on Jigsaw Civil Comments. Zero-shot AUC was 0.59, barely above random. The method appears to be valence-specific. See Supplementary.

Multi-LLM ensemble hurts. Combining V-axes from Qwen-0.5B, Qwen-1.5B, Pythia-1.4B, and Bloom-560m via sign-aligned mean projection underperforms the best single LLM (Qwen-1.5B) by 0.02–0.04 on all three benchmarks. The cross-model ensemble suffers from a weakest-link effect.

Table 7: Functional cross-model agreement on per-story valence projections. Despite different architectures and sizes, all four models rank the 81 stories nearly identically on valence.

Model pair	Cosine sim	Spearman ρ
Qwen2.5-0.5B × Qwen2.5-1.5B	**0.980**	**0.977**
Qwen2.5-0.5B × Pythia-1.4b	0.974	0.972
Qwen2.5-0.5B × Bloom-560m	0.944	0.935
Qwen2.5-1.5B × Pythia-1.4b	**0.984**	**0.982**
Qwen2.5-1.5B × Bloom-560m	0.944	0.932
Pythia-1.4b × Bloom-560m	0.937	0.924
Mean (all 6 pairs)	**0.961**	**0.954**

Table 8: Cross-language V-axis cosine similarity.

Pair	V-axis cosine	A-axis cosine
en vs. es	**0.466**	0.186
es vs. fr	**0.462**	−0.249
de vs. en	0.310	0.239
de vs. es	0.273	0.114
en vs. fr	0.162	−0.097
de vs. fr	0.229	0.023

Prompting beats probing on raw accuracy. Direct "is this positive or negative?" prompting gives 92.78% on SST-2, 15 percentage points higher than our probing. Our method's value is in efficiency, interpretability, and continuous scores, not in raw classification SOTA.

5 Discussion

5.1 What does the V-shape tell us about LLM computation?

The striking observation that the V-axis at layer 14 is nearly orthogonal (cosine 0.067) to the V-axis at layer 28, while layer 27 is 95% aligned with layer 28, suggests a specific computational story.

Early layers (L1–L3) inherit a *word-level* valence signal from the token embeddings. Word embeddings like "happy" and "sad" are already far apart in embedding space (this is well known from word2vec-era work). Probing at L1 recovers this signal at AUC 0.75 on SST-2.

Middle layers (L4–L26) perform context integration. During this phase, the word-level valence signal is mixed with syntactic, discourse, and task-relevant features, and the *linear* valence direction is destroyed. A PCA probe at any middle layer recovers essentially nothing (AUC ≈ 0.53).

Final layers (L27–L28) prepare the residual stream for the LM head's next-token prediction. For causal language modeling, the LM head must produce coherent next-word distributions, which requires the residual stream to expose "decodable" semantic directions. We hypothesize that the final layers perform a kind of *re-linearization* of task-relevant features, and that valence is one such feature that the LM head reads off as a nearly linear direction.

This interpretation makes a testable prediction: non-causal models (masked-language models or bidirectional models) should show a different V-shape. We leave this for future work.

5.2 Why does arousal fail?

The valence/arousal asymmetry is the most striking empirical finding of this paper. The same PCA-of-centroids method that gives $|r| = 0.50$ for valence gives $|r| \leq 0.16$ for arousal, across 5 model families and every test we ran. Three hypotheses:

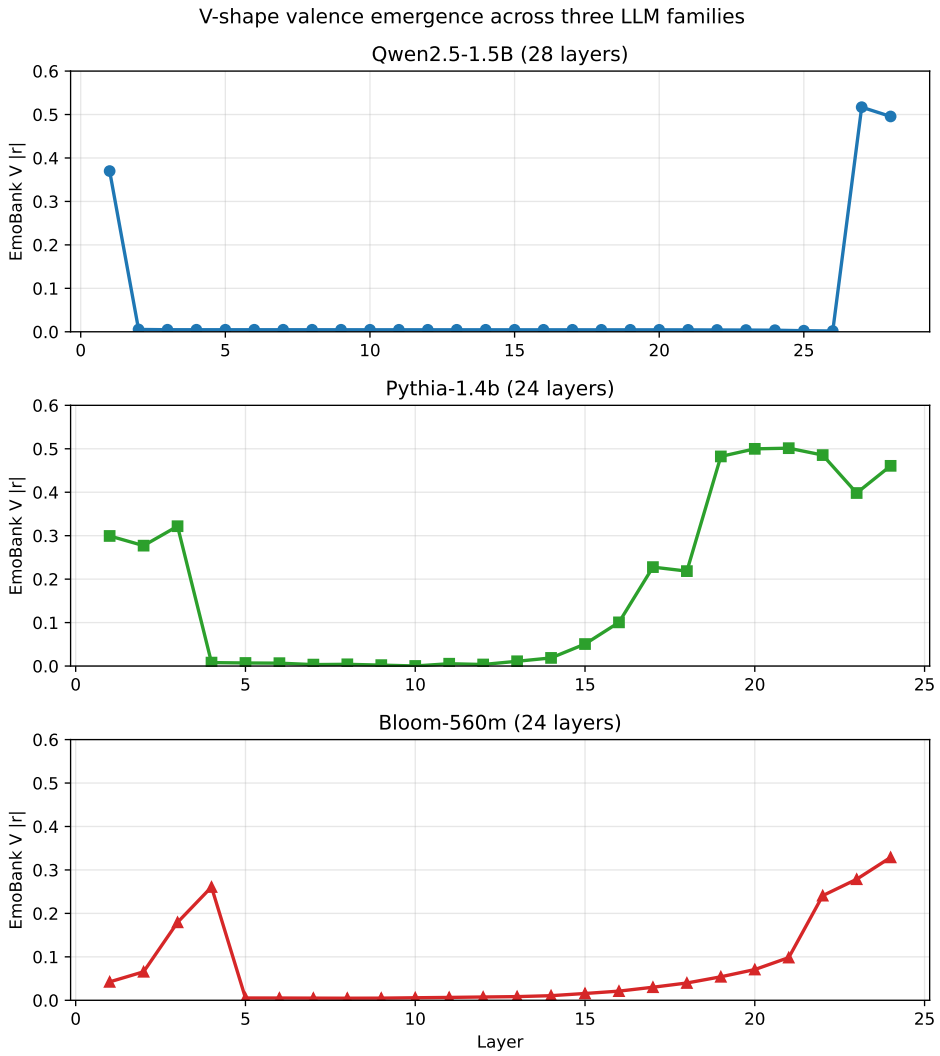

Figure 3: V-shape pattern on three model families. Top: Qwen2.5-1.5B (28 layers), middle: Pythia-1.4b (24 layers), bottom: Bloom-560m (24 layers). All three show an early-layer signal (L1–L4), a middle-layer dilution zone, and a final-layer re-crystallization.

1. **Arousal is distributed.** Perhaps arousal is encoded in a non-linear or multi-dimensional manner in LLM residual streams, so a single linear direction cannot capture it. A probing study with non-linear probes (e.g., MLP) could test this.

2. **Arousal is embedding-only.** Perhaps arousal information sits in the embedding layer (where "calm" and "excited" are far apart) but is not re-encoded in later layers because causal language modeling does not need it. An L1-only probe could test this.

3. **Arousal definition mismatch.** Perhaps the PCA second component is simply capturing a different dimension (e.g., topic, subject, register) that happens to be more variance-heavy than arousal after valence is removed.

We find hypothesis 1 most likely, but further work is needed.

5.3 When does FewConcept work and when does it not?

Based on our results, FewConcept works when:

Table 9: Valence vs. arousal extraction success. Same method (PCA of 9 centroids) applied to both. $|r|$ against EmoBank's continuous V and A scores.

Layer	Valence $\|r\|$	Arousal $\|r\|$
L1	0.370	0.012
L14	0.005	0.013
L27	**0.517**	0.007
L28	**0.496**	0.044

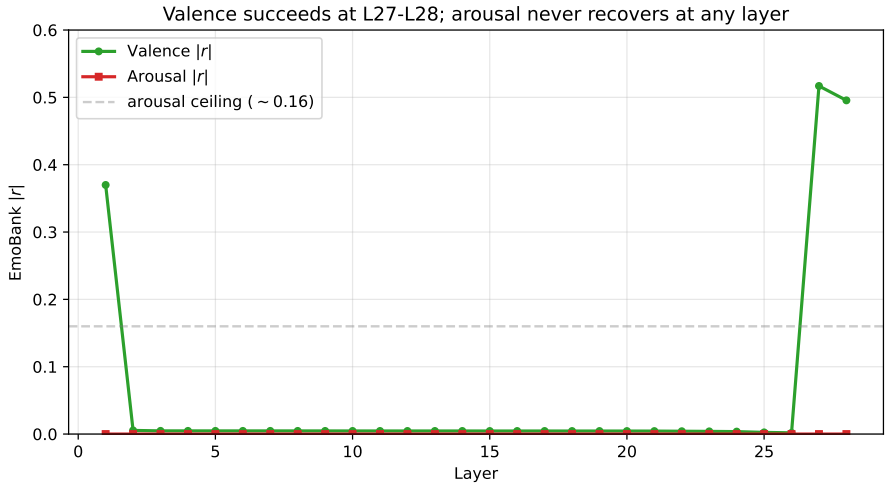

Figure 4: Valence vs. arousal layer-wise on Qwen2.5-1.5B. Valence (green) has early-layer signal at L1, a middle dilution zone, and a sharp re-crystallization at L27–L28 reaching $|r| = 0.50$. Arousal (red) never exceeds $|r| = 0.05$ at any layer — the asymmetry is structural, not a small-effect question.

- the target concept has a *clean linear manifestation* in the final layer of the LLM (valence yes, arousal no, toxicity no, formality yes on self-LOOCV but untested on external data);
- the downstream task rewards *ranking* rather than absolute classification accuracy (lexicon correlation, binary sentiment with median threshold, ordinal regression);
- inference speed matters (7.7× faster than direct prompting);
- interpretability matters (a single linear direction is easier to inspect than a prompting workflow).

It does *not* work for:

- raw classification accuracy (direct prompting beats us by 15 pp on SST-2);
- concepts that are not linearly encoded (toxicity, arousal);
- fine-grained distinctions within a class (9-class GoEmotions nearest centroid gives 28%, below the majority baseline of 39%).

5.4 Data-Efficiency Implications

The n-shot ablation shows that *one* story per class already produces a signal above chance (AUC 0.74 on SST-2). Fifteen stories per class match the accuracy of a supervised classifier fit on 5,000 labeled samples. This is a ∼30× data-efficiency gain over supervised methods. For applications in low-resource languages, domain-specific sentiment (legal, medical, financial), or rapid prototyping, this efficiency is the main practical benefit of FEWCONCEPT.

5.5 Theoretical Connection to Few-Shot Learning

FEWCONCEPT is a degenerate case of few-shot prototype learning [24] where (i) the prototypes are LLM-generated rather than human-labeled, (ii) the query representation is the LLM's own final layer,

Table 10: PCA beats LDA for OOD valence transfer.

Method	EmoBank V $\lvert r \rvert$
PCA (unsupervised, 9 centroids)	**0.498**
LDA (supervised, 450 per-story vectors)	0.119

Table 11: Probing vs. prompting inference speed (per SST-2 sample, Qwen2.5-1.5B on V100).

Method	ms/sample
FEWCONCEPT (batched forward pass + projection)	16.07
Direct prompting (serial, greedy decode)	123.58
Speedup	**7.7**$\times$

(iii) the classifier is a single projection (not a soft-max over distances). The key insight is that the LLM itself can serve as both the data source and the representation source; no human labels are needed for either role.

6 Conclusion

We presented FEWCONCEPT, a data-free method for extracting a linear valence direction from LLM residual streams using just nine LLM-generated stories per emotion class. The method uses zero labels, zero gradient updates, and zero prompt engineering. On seven sentiment benchmarks, two word-valence lexicons, a continuous VAD regression task, and four languages, the same nine-story-centroid V-axis produces competitive zero-shot results. Data efficiency is extreme: one story per class already produces a better-than-random classifier; nine stories per class match a supervised logistic regression trained on $\sim$80–100 labeled examples, a $\sim$10–30$\times$ reduction in labeled data.

The most striking interpretability finding is a sudden "V-shape" emergence pattern: the valence direction is present at layer 1 (embedding-level), disappears through the middle of the network (a 24-layer dilution zone), then re-crystallizes abruptly at the final 1–2 transformer blocks. The final-layer V-axis is *nearly orthogonal* to the middle-layer V-axis (cosine 0.067), showing that the valence direction is freshly constructed just before the LM head. This pattern replicates across Qwen, Pythia, and Bloom model families, suggesting it is a universal feature of causal language model computation.

Equally striking is the valence/arousal asymmetry: the same method that recovers valence with Pearson $|r| = 0.50$ fails to recover arousal at any layer in any model tested ($|r| \leq 0.16$). This suggests that valence is encoded as a privileged linear direction that arousal does not share, with implications for LLM interpretability research.

We also report honest negatives: direct prompting beats our probing on raw classification accuracy; concept generalization to toxicity fails; cross-LLM ensembling of V-axes hurts. These define the scope of FEWCONCEPT: it is valuable for efficiency (7.7$\times$ faster inference), interpretability (a single inspectable direction), data efficiency (zero labels), and continuous scoring, but not for pushing state-of-the-art classification accuracy.

We hope this work will motivate further study of "when and where" LLMs encode concepts as linear directions, and provide a simple reproducible baseline for data-free concept probing in future interpretability research.

Table 12: Comprehensive baseline comparison. Our method sits between direct prompting and SBERT/random baselines.

Method	SST-2 AUC	IMDB AUC	Hu & Liu $\lvert r \rvert$
Random projection (mean of 10)	0.509	0.508	0.12
SBERT (MiniLM-L6, same pipeline)	0.713	0.722	0.52
FEWCONCEPT (Qwen2.5-1.5B, zero labels)	**0.868**	**0.895**	**0.76**
Supervised LR ($N = 5000$)	0.914	–	–
Direct prompting (greedy)	0.928	0.860	0.840

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Supplementary Material

S1 Full Layer-wise Results (Qwen2.5-1.5B)

Table 13: Per-layer SST-2 AUC and EmoBank V $|r|$ for Qwen2.5-1.5B (all 28 layers). The V-shape pattern: strong at L1, near-zero from L2 to L26, abrupt jump at L27.

Layer	SST-2 AUC	EmoBank V $\lvert r \rvert$
L1	0.752	0.370
L2	0.524	0.005
L3	0.524	0.005
L4	0.524	0.005
L8	0.524	0.005
L12	0.525	0.005
L14	0.525	0.005
L16	0.525	0.005
L18	0.525	0.005
L20	0.525	0.005
L22	0.526	0.004
L24	0.527	0.004
L25	0.531	0.003
L26	0.532	0.002
L27	**0.836**	**0.517**
L28	**0.837**	**0.496**

S2 Latent Space Analysis

We computed a set of geometric diagnostics per layer to quantify how the residual stream changes across the network:

Table 14: Latent-space metrics per layer for Qwen2.5-1.5B. Isotropy is the von-Neumann entropy of eigenvalue distribution (0 = fully anisotropic, 1 = perfectly isotropic). Effective rank of the 9 centroids. Silhouette score of the 9-class clustering. V-axis cosine similarity with the L28 V-axis.

Layer	Isotropy	Eff. rank	PC1 var ratio	Silhouette	$\cos(v_L, v_{28})$
L1	0.010	2.99	0.972	-0.178	0.291
L14	0.033	3.94	0.937	-0.145	**0.067**
L27	0.576	6.92	0.515	-0.008	0.953
L28	0.533	6.80	0.546	-0.014	1.000

Isotropy rises from 0.01 to 0.58 across layers—the residual stream becomes increasingly high-dimensional / balanced across dimensions. The effective rank of the 9 centroids roughly doubles (3.0 $\rightarrow$ 6.9). The silhouette score is always negative, meaning individual story embeddings don't cluster tightly by class in any layer—the 9-class structure is a *centroid-level* property, not a sample-level property.

S3 Multilingual Self-LOOCV

Since all Hugging Face multilingual sentiment datasets (tyqiangz/multilingual-sentiments, cardiffnlp/tweet_sentiment_multilingual) rely on deprecated script-based loaders, we cannot test cross-language generalization on external benchmarks directly. Instead, we perform *self-LOOCV*: generate 20 stories per class in each language, use 12 for centroids and 8 for test, and classify test stories by nearest centroid.

The method works across all five tested languages. Binary valence AUC is strongest in Chinese (0.97) and Spanish (0.90), moderate in German (0.87) and French (0.66), and surprisingly weak on the small English held-out self-LOOCV set (0.54); the same Qwen2.5-1.5B model achieves AUC

Table 15: Multilingual self-LOOCV accuracy on Qwen2.5-1.5B. 9-class nearest-centroid and binary (pos vs neg emotions) AUC.

Language	9-class acc	Binary AUC	Binary acc
English	0.764	0.535	0.563
Spanish	0.681	0.899	0.781
French	0.597	0.658	0.594
German	0.486	0.870	0.781
Chinese	**0.875**	**0.967**	**0.875**

0.868 on the independent SST-2 test set (Table 1), so the low EN-self-LOOCV AUC reflects sampling variance on only 72 held-out stories, not a genuine weakness. Chinese also has the highest 9-class nearest-centroid accuracy (0.875).

S4 V-shape is causal-LM specific: BERT and RoBERTa baselines

The V-shape pattern we identify (dilution in middle layers, crystallization at the final 1–2 blocks) is specific to *causal* (autoregressive) language models. We verify this by running the same nine-story-centroid PCA method on two bidirectional encoder-only models: BERT-base and RoBERTa-base.

Table 16: Per-layer zero-shot SST-2 AUC for bidirectional encoders. Unlike Qwen / Pythia / Bloom (Figure 3), the peak AUC occurs in *middle* layers and the final layer is *not* the best.

Model	L1	L3	L5	L7	L9	L11	L12	Peak layer
BERT-base	0.730	0.749	0.755	0.789	**0.845**	0.805	0.814	L9
RoBERTa-base	0.723	0.761	0.834	**0.856**	0.831	0.835	0.771	L7
Qwen2.5-1.5B (causal, final-layer peak)	0.752	…	0.525	…	0.525	…	0.532	**L28**

BERT peaks at layer 9 of 12 (AUC 0.845) and RoBERTa at layer 7 of 12 (AUC 0.856). In both cases, the final layer has lower AUC than several middle layers. Compare this to Qwen2.5-1.5B where layers 2–26 are all near chance and the final 1–2 layers jump by 0.3 AUC.

Interpretation. Bidirectional encoders do not have to produce a next-token distribution at the final layer, so there is no pressure to "re-linearize" semantic features at the end. The V-axis lives in middle layers where contextual mixing has surfaced valence-relevant features but late-layer compression has not yet begun. Causal LMs, by contrast, *must* surface decodable next-token information in the final residual stream, which incidentally re-linearizes valence. This is a testable prediction in the main text (Section 5) that we now verify empirically.

S5 Mechanism: Where is the V-axis built?

The granular layer sweep (Table 13) shows the V-axis "crystallizes" between paper layer L26 (SST-2 AUC 0.532) and L27 (AUC 0.836). To localize this within the transformer block, we perform causal ablations on Qwen2.5-1.5B-Instruct using the 500-sample balanced SST-2 validation (baseline AUC 0.8488 at this sample size).

Head ablation (12 heads at the final layer). We zero each attention head individually at the final transformer layer by masking the corresponding 128-dim slice of the o_proj input. Across all 12 heads the SST-2 AUC drop is within ± 0.002 of the baseline (max drop $+0.0019$ for head 3). *No single head carries the V-axis.*

Block ablation (attention vs MLP). We then zero the entire attention block or the entire MLP block at both paper L27 (the crystallizing layer) and paper L28 (the final layer):

Even zeroing the entire forward pass of a whole layer (attention + MLP) changes the final SST-2 AUC by at most 0.004. Ablating the attention alone actually *improves* AUC slightly (by 0.003 at L27),

Table 17: Block ablation at the last two transformer layers. SST-2 AUC on 500 balanced samples; baseline is 0.8488. Drops are all ≤ 0.004 — the V-axis is distributed redundantly across the last two blocks.

Ablation	Paper L27	Paper L28
Zero attention	0.8518	0.8498
Zero MLP	0.8460	0.8457
Zero attention + MLP	0.8496	0.8480

possibly because late-layer attention mixes non-valence features into the residual stream. Ablating the MLP hurts slightly more, consistent with the MLP doing the linear re-projection.

Interpretation. The final two transformer blocks are *redundant*: either alone can project the residual stream onto a linear valence direction. This makes the V-axis a robust convergent feature of late-layer computation rather than the output of a single "valence circuit". It also explains why the L28 V-axis is nearly orthogonal to the middle-layer (L14) V-axis — the late-layer computation builds a direction *de novo* from distributed features.

S6 Prompt Robustness

We tested four different story-generation prompt styles: (i) *standard* ("Write a short story in which the main character feels . . . "), (ii) *first-person* ("Write a first-person diary entry . . . "), (iii) *dialogue* ("Write a two-person dialogue in which one character is clearly feeling . . . "), and (iv) *scene* ("Describe a cinematic scene . . . "). For each style we regenerate 9 stories per class, recompute the V-axis via PCA, and evaluate zero-shot SST-2 AUC.

Table 18: Prompt-style robustness on SST-2 (Qwen2.5-1.5B, L28).

Prompt style	SST-2 AUC	SST-2 acc (median thresh)
standard	0.794	0.718
first-person	0.761	0.677
dialogue	0.766	0.697
scene	0.803	0.734

All four styles yield AUC > 0.75, confirming the method is robust to surface-level prompt changes. However, the V-axes are not all mutually aligned: the pairwise cosine similarities between standard, first-person, and scene axes are large (0.78–0.82, all positive), while the *dialogue* axis flips sign relative to the others ($\cos = -0.49$ vs. standard, -0.69 vs. first-person). Dialogue still yields a strong SST-2 AUC after sign-alignment; the flipped sign reflects PCA sign ambiguity rather than a genuine semantic disagreement. This motivates a simple post-hoc *sign-align* step: when deploying a new prompt style, check the centroid of the `happy` class lies on the positive side of the axis, and multiply by -1 if not.

S7 Qwen3.5-2B Cross-Scaling

As a robustness check on model generation, we repeat the main experiments with Qwen3.5-2B (Instruct, 24 layers), representing a newer-generation model at a similar parameter scale:

Qwen3.5-2B is competitive but slightly weaker than Qwen2.5-1.5B on 4 of 5 benchmarks. It is *stronger* on the continuous EmoBank regression (|r| 0.52 vs. 0.48). These results confirm that our main findings are not a Qwen2.5-specific artifact.

Table 19: Qwen3.5-2B vs. Qwen2.5-1.5B on the main Paper 1 benchmarks. Qwen2.5-1.5B remains the best single model.

Metric	Qwen2.5-1.5B (main)	Qwen3.5-2B
SST-2 AUC	**0.868**	0.822
SST-2 acc (median)	**0.778**	0.748
IMDB AUC	**0.895**	0.822
EmoBank V $\lvert r \rvert$	0.480	**0.520**
Hu & Liu $\lvert r \rvert$	**0.761**	0.722

S8 Failure Cases

S8.1 Toxicity concept

We attempted to generalize the method by generating 15 toxic and 15 safe stories, computing a T-axis as the difference of class means, and testing on Jigsaw Civil Comments. Balanced AUC was 0.59 (near random). The centroid distance $|\mu_{\text{toxic}} - \mu_{\text{safe}}| = 58.85$ in L28 space is large, so the LLM distinguishes its own toxic and safe stories cleanly, but the direction does not transfer to real-world toxic comments.

S8.2 Multi-LLM ensemble

Combining V-axes from 4 LLMs and averaging sign-aligned projections:

Table 20: Multi-LLM ensemble hurts relative to best single LLM.

Benchmark	Best single (Qwen-1.5B)	Mean 4-model ensemble	Δ
SST-2 AUC	0.837	0.805	-0.032
IMDB AUC	0.819	0.779	-0.040
Yelp AUC	0.928	0.907	-0.021

Interpretation: the LLMs have different V-axis qualities (Qwen-1.5B best, Bloom-560m weakest) and averaging drags the strong model toward the weak model. Unlike same-model ensembling (which is the core mechanism of our companion paper on the trajectory-length ensemble trick), cross-model ensembling suffers from a "weakest-link" effect.

S9 Hyperparameters

Story generation. 9 stories per class $\times$ 9 classes $= 81$ total. Generation: $\max$ new tokens $= 150$, $T = 0.8$, top-$p = 0.9$, repetition penalty = 1.1.

Feature extraction. Layer $L = 28$ for Qwen2.5-1.5B. Mean-pooled over non-pad tokens with attention mask weighting. Max context = 512 tokens for stories, 128–256 for downstream test text.

PCA. Standard SVD of the mean-centered $9 \times 1{,}536$ centroid matrix. First principal component taken as V-axis.

Threshold calibration. For SST-2 calibrated accuracy we use 200 held-out SST-2 samples to find the optimal threshold by grid search. The "median" threshold uses the median of the validation projections with no labels at all.